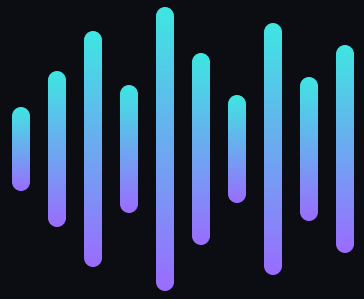
CASE STUDY · CREATIVE TECHNOLOGY

Building a Privacy-Conscious AI Music Metadata Agent

From a personal cataloging problem to a working private beta deployed through Replit.

Jamil Houston

Product • AI • Security • Deployment



01 · THE OPPORTUNITY

Metadata matters—but tagging is still slow and inconsistent.

Creators and catalog owners often repeat the same subjective work across disconnected tools.

FRICTION 01

Repetitive work

Genre, mood, instrumentation, themes, and use cases are entered track by track.

FRICTION 02

Inconsistent language

Different listeners describe the same recording with different levels of detail.

FRICTION 03

Fragmented delivery

Useful metadata must be reformatted for catalogs, collaborators, and exports.

The goal was not to replace judgment. It was to create a stronger first draft.

02 · THE PRODUCT

A faster path from audio to usable metadata.

Users submit a track, review editable AI suggestions, and export the approved result.

1 · Upload or link
Upload an audio file or provide a supported Spotify or Apple Music link.

→

2 · Analyze
Evaluate technical traits and listening-based characteristics.

→

3 · Review
Edit genre, mood, instruments, themes, energy, tempo, and sync use.

→

4 · Export
Copy text or create Excel, Google Sheets, PDF, and Word outputs.

Genre · Mood · Instrumentation · Themes · Energy · Tempo · Sync Use

03 · HOW IT WORKS

Two kinds of evidence. One reviewable result.

Streamlit interface
Collects the audio or supported link and optional track context.

→

Audio analysis
Combines measurable characteristics with listening observations.

→

Structured classification
Maps evidence into consistent metadata categories.

→

Human review
The user edits, approves, and exports the final metadata.

Machine assistance
Accelerates listening, organization, and the first-pass vocabulary.

Human authority
Controls meaning, accuracy, context, and the final catalog record.

04 · PRIVACY BY DESIGN

Collect less. Retain less. Invent nothing.

- ✓ No consumer account or contact information required
- ✓ Temporary audio processing with deletion after analysis
- ✓ Permission-to-upload acknowledgment before processing
- ✓ No persistent catalog saving for anonymous visitors
- ✓ No fabricated ownership, rights, or identifier data

Minimum necessary

Only the information required to analyze and return a result should enter the workflow.

Privacy is reflected in the product behavior—not only in documentation.

05 · SECURITY AND RESPONSIBLE USE

Layered controls protect users, credentials, and cost.

Credential safety

Server-side project credentials, restricted permissions, rotation, and secrets kept outside the codebase.

Controlled access

Private-beta access, daily usage limits, and spending controls reduce misuse and surprise costs.

Upload boundaries

File-size, format, and track-length limits reduce risk and keep processing predictable.

CODE	API	USAGE	BUDGET
Protected ignored files	Project-scoped access	Daily analysis cap	Spending controls

06 · BUILD AND DEBUG

A successful listening pass still exposed a schema problem.

THE STACK

Python | Streamlit | OpenAI audio intelligence | Structured classification | Pydantic validation | Export libraries

THE FAILURE

Strict JSON schema rejected a flexible output field.

The audio analysis completed, but the structured result could not pass validation.

1 · Isolate
Confirmed the failure occurred after the listening pass.

→

2 · Diagnose
Traced the issue to strict structured-output compatibility.

→

3 · Configure
Adjusted the validated output schema intentionally.

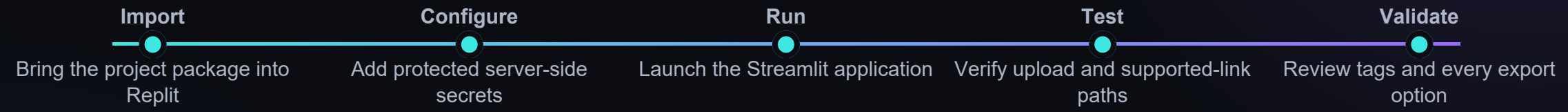
→

4 · Verify
Reran tests and the complete user journey.

07 · DEPLOYMENT

From project package to protected Replit application.

Deployment was validated as a complete user journey—not just a successful startup.



DELIVERED EXPERIENCE

Accessible private beta

The working application is presented through the Labs case study on jamilhouston.com.

A portfolio artifact that demonstrates end-to-end product thinking.

WHAT IT PROVES

- Applied AI grounded in a real workflow
- Privacy and security considered before launch
- Practical debugging and end-to-end testing
- Deployment and portfolio communication

WHAT COMES NEXT

- Per-user authentication and rate limits
- Improved monitoring and operational visibility
- Exact repository or public case-study link
- Evaluate mobile packaging when the web beta is stable

> Build around a real need. Protect the user. Keep the human in control.